

Finnish company ZenRobotics has come up with the world's first robotic waste sorting system. It employs AI for smart recycling of waste materials. Using computer vision, machine learning and AI, the robot can sort and pick recycled materials from moving conveyor belts.

e-Waste management in India

The government of India has issued revised E-Waste Management Rules 2016, which came into force in October 2016. The Pollution Control Board of India has suggested different colours for different dustbins, namely, green for biodegradable waste, black for e-waste, and blue for plastics and metals. But these dustbins are not used properly due to either ignorance or lack of seriousness of people. Black dustbins for dumping e-waste are rarely found on the streets. So, until now, most of the garbage and e-waste from Indian cities are in landfills or have gone down the rabbit hole.

Absence of strict regulations on e-waste management, lack of awareness among the masses and absence of efficient infrastructure are all leading to the unchecked spread of e-waste in the country. Lack of proper procedures for the collection, disposal and recycling of e-waste is another major reason for the poor state of



Figure 4: Colour-coded dustbins for use in India

e-waste management in India.

India still lags behind other countries when it comes to an automated waste management system. Recently, many Indian companies have started working on e-waste management and recycling solutions. These include Namo e-Waste, Revive Electronic Waste, Attero Electronics, Let's Recycl, Karo Sambhav, Cerebra and E-Parisaraa.

India ranks third in the world in terms of high-quality research publications in AI. However, currently, implementations of AI and machine learning in e-waste management on a large scale are hardly seen here. Current e-waste management systems are unable to efficiently deal with the massive amounts of e-waste generated every day. Automating the processes of sorting and disposal using AI and machine learning would help in handling e-waste more effectively. It is time to switch to AI and

machine learning for smart recycling and e-waste management in India.

To sum up

Our lack of initiative in e-waste management has

resulted in many recyclable precious items being thrown into dustbins, which have ended up in landfills. It is the need of the hour to reduce, reuse and recycle e-waste. Gold, silver and other precious metals can be obtained by recycling e-waste. Non-recyclable items can either be sent to landfills or burned (incinerated) under controlled environments.

We can expect a significant decrease in e-waste if traditional techniques are replaced with automated intelligent solutions and sorting AI-robots. As sensible individuals, we should reduce e-waste and make others aware of its value, in addition to its harmful effects. This is important to save our lives and the environment. **END** 🐧

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By: Sani Theo

The author is a technical editor at EFY.

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MONTH	THEME
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May 2019	AI, Deep Learning and Machine Learning
June 2019	DevOps Special
July 2019	Blockchain and Open Source
August 2019	Database Management and Optimisation
September 2019	Open Source and IoT
October 2019	Cloud Special: BigData, Hadoop, PaaS, SaaS, IaaS and Cloud
November 2019	Open Source on Windows
December 2019	Open Source Programming (Languages and Tools)
January 2020	Data Security, Storage and Backup
February 2020	Best in the World of Open Source (Tools and Services)